

Amazon Books Inferences

A Report on Exploratory Data Analysis, Modeling and
Validating using Machine Learning Techniques

Group 22

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1 Introduction

In the modern digital era, the explosive growth of e-commerce platforms and digital libraries has fundamentally changed how consumers discover and consume books. Such e-commerce platforms have generated vast amounts of data that can be leveraged to understand consumer behavior, product performance, and market trends. Among these platforms, Amazon is one of the world's largest online marketplaces, offering millions of products across various categories, including books. The availability of those data provides valuable opportunities for analysis and predictive modeling. With this abundance of choices comes the challenge of information overload: users are often overwhelmed by the sheer volume of options, making it difficult to discover books aligned with their specific tastes.

To address this challenge, data-driven recommendation systems and predictive analytics have become critical components of user engagement and business strategy. Using historical user interaction data, such as explicit ratings, reviews, and demographic information, machine learning algorithms can uncover latent patterns in reading habits, personalize content delivery. Through this project, data-driven methods are applied to transform raw book data into actionable knowledge, demonstrating how analytical techniques can be used to better understand consumer preferences and market dynamics within the online book marketplace.

2 Research Questions

1. **How can user reading behavior be used to recommend books to readers in the Amazon Books dataset?**

By identifying users with similar preferences, the system aims to recommend books that are likely to match a user's interests.

2. **Which book and user-related features help predict whether a book will receive a high rating?**

This question focuses on whether some features are related to high ratings. It matters because understanding these factors can show what influences book popularity and reader satisfaction.

3. **Can we use user- and book-related features to predict whether a book will receive a high rating?**

By answering the question, we can understand the reading preferences, recommend books that meet the interests of the users, and predict whether a user is likely to enjoy a book. This can improve the user experience and help platforms provide more personalized recommendations.

3 Data Description

The data set used in this project is the Amazon Books data set from Kaggle. It contains three main tables: books, ratings, and users. The unit of analysis is mainly a user-book interaction, where each row in the ratings table represents one user's interaction with one book.

The original book data set contains 271,379 rows and 8 columns, including ISBN, book title, book author, year of publication, publisher, and 3 columns containing links to the image of the cover of each book. The rating data set contains 1,149,780 rows and 3 columns, including user ID, ISBN, and book rating. The user data set contains 278,858 rows and 3 columns, including id, age, and country.

3.1 Dataset Preprocessing

Several preprocessing steps were applied to clean and prepare the Amazon Books data set. For the book data set, unnecessary image URL columns were removed because they were not useful

for the analysis or modelling process.

The user data set required additional preprocessing because the location variable contained inconsistent and unstandardised country names. To address this issue, a two-step validation method was used. First, the country names in the data set were compared with the official country and territory names available in the Python `pycountry` library. Second, countries with more than 50 users were manually checked to identify names that were not matched in the first step. Some common alternative names and abbreviations were then manually added to the valid country list. For example, “usa” was assigned to “united states.”

After this cleaning process, about 98% of the users of the original dataset were preserved and assigned to standardised country names. The remaining 2% were removed from the set because their location values were too unstandardised or unclear. The Location column was dropped and instead the column Country was used for location-wise analysis. Two countries with more than 50 users, “españa” and “yugoslavia,” were not included because the majority of users used the standardised name “spain”, and “yugoslavia” no longer exist as a country.

The ratings, users, and books datasets were then merged into one main dataframe using User-ID and ISBN as key variables. To reduce noise and simplify the modelling process, the data was restricted to books from the top 10 most common publishers.

Several new variables were created to support the analysis. A book-read-count variable was created to show how many times each book appeared in the dataset. An author-read-count variable was also created to measure author popularity. For the prediction model, ratings equal to 0 were removed because they represent implicit interactions rather than actual rating scores. After removing these values, a book-rate-count variable was created to count the number of explicit ratings for each book.

Missing values in the Age variable were handled by replacing them with the median age. This allowed the age variable to be kept in the prediction model without removing a large number of observations. Finally, the rating variable was transformed into a binary target variable called rating-over-k. Ratings greater than or equal to 8 were labelled as 1, meaning a high rating, while ratings below 8 were labelled as 0. This prepared the data set for a classification task in which the goal is to predict whether a user will give a book a high rating.

4 Exploratory Data Analysis

Some of the questions that first appeared could be answered with EDA alone. In this section, we explore these questions, provide answers, and dive deeper into the dataset.

4.1 Time Periods Related EDA

Right after we got acquainted with the dataset, we noticed many missing age values in the user table. An idea we thought about was to predict the missing ages using the available information. For this purpose, we had to answer a couple of questions, using EDA, to see if it was plausible.

One of those initial questions was “Is there any correlation between the user age and the books they tend to read?” In this subsection, we answer this question.

First, we started by calculating the books read per user per century. A note worth mentioning here is that books from only the 14th, 19th, 20th, and 21st century were actually read from users in our dataset. After calculating the numbers, we then computed the correlation coefficients across all columns. Although one might anticipate that the User-ID has no correlation with any century or age, it is nice to cross-check this.

The correlation coefficients that would answer our initial question are those located in the Age row for each century column in Table 1. As we can see, all of them are very close to 0, showing that there is no significant correlation between those two factors.

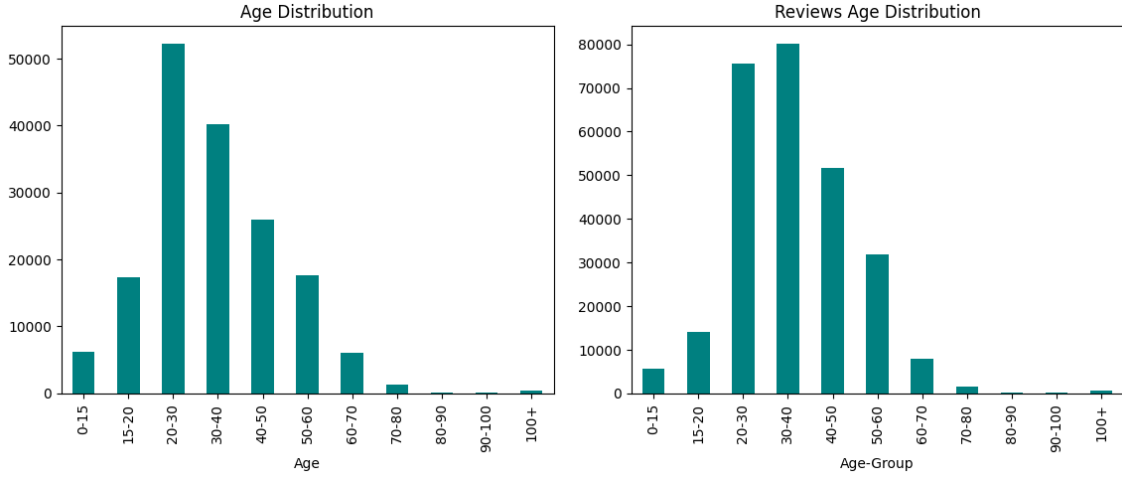
On the other hand, we notice a quite firm correlation between the books from the 20th and 21st century that users read. This is a good indicator that a book recommendation system could be explored. This will be discussed in Section 5.

	User-ID	1300s	1800s	1900s	2000s	Age
User-ID	1.000000	0.000222	-0.001531	0.002517	-0.000312	-0.004814
1300s	0.000222	1.000000	-0.000022	0.017870	0.021033	-0.003247
1800s	-0.001531	-0.000022	1.000000	0.135497	0.053382	0.005912
1900s	0.002517	0.017870	0.135497	1.000000	0.672872	0.015324
2000s	-0.000312	0.021033	0.053382	0.672872	1.000000	0.013847
Age	-0.004814	-0.003247	0.005912	0.015324	0.013847	1.000000

Table 1: Correlation matrix for User-ID, Centuries and User Age

4.2 Age Related EDA

In Figures 1a and 1b we have visualised the Age distribution of the users in the data set, and how many book ratings were produced by users in the given age groups. As we can see, the highest percentage of users in our dataset are between 20-40 years old, with a higher proportion of the 20-30's. On the other hand, we see that a higher proportion of the 30-40 year olds place more reviews than the 20-30 year olds. Note that the distribution of users who have purchased a book and have not left a review is identical to that of Figure 1b, and therefore we have left it out.



(a) Distribution of Age across users in the dataset

(b) Distribution of reviews per Age across users

Further analysis on age contributed to the rejection of developing a predictive model for the same parameter. This comes from the fact that in Figure 2 we see no clear distinction between the ratings between different age groups. An exclusion to this is the 80-90 age; however, the results could be inconclusive, since this group is much smaller in sample size compared to any other group.

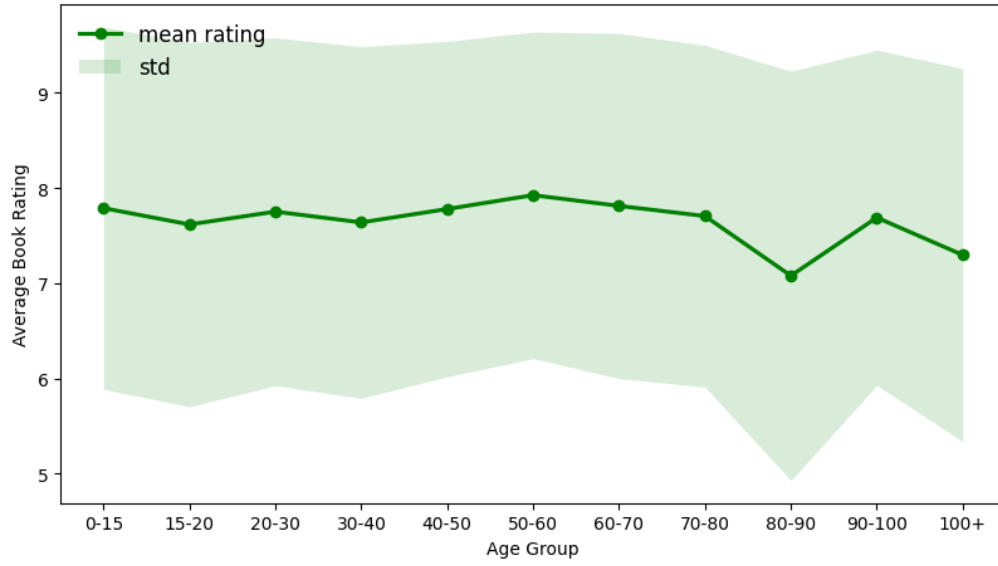


Figure 2: Average Book Rating per Age Group

4.3 Country Related EDA

One of the first questions that appeared, regarded the average review scores per country. We were curious "Do average ratings per country vary considerably between each other?"

In Figure 3 we have visualised the distribution of book purchases per country. Recall that book purchases here include all ratings 0-10, where 0 is the lack of one, and 1 is the lowest actual rating. Therefore, the scale of this graph is rather high and possesses un insightful further characteristics for analysis, due to the mix of these two aspects.

Therefore, we decided to divide all purchases into rated and non-rated purchases, both of which are visually represented in Figure 15 and, respectively, Figure 16 in Appendix A.



Figure 3: Sample Sizes of Book Purchases per Country

Due to the uneven split of reviews per country in the data set, we decided to leave countries with fewer than 100 reviews out of the statistical calculations and the overall category from the

modelling afterward. We decided that 100 reviews would be a good splitting rule, since under Central Limit Theorem, samples with above 30 observations allow the sample mean to be treated as approximately normally distributed, allowing for a more reliable estimate of the means. We note that statistical precision still varies considerably within this filtered set, due to over-saturation in some countries compared to others (USA data is more than 500 times more than Romania in this case, for example). For transparency, we have included the visualisation with all countries, disregarding the sample size in Appendix A, Figure 14.

In Figure 4 we have visualised the mean book ratings per country, where we can see that the majority falls between 7 and 9, with average mean of all of about 7.6 amongst all countries. Of all countries, Romania, with a total of about 450 reviews left, has the highest rating of about 9.2 and the smallest standard deviation of approximately 1.3. The lowest average rating is for Israel, about 6.7, with a standard deviation of about 2.2, in approximately 120 reviews.

In Figure 5, we computed the standard deviation of each country's rating, where we see that the average one across all is about 1.8. Considering this information, we find that for a randomly selected review we can say that the score might be roughly between 6 and 9.5 (mean $\pm$ std). As one can imagine, on the scale 1-10, this suggests that happy customers may tend to leave reviews more often compared to unhappy ones, making it a reasonable hypothesis for investigating a positive bias.



Figure 4: Average Book Rating per Country for Countries with more than 100 reviews



Figure 5: Standard Deviation Book Rating per Country for Countries with more than 100 reviews

5 The Model

This section covers the recommendation system that creates a subset of books and a corresponding "vote score" in relation to a set of seed books; and the classification models we used in order to answer our research questions. In particular, we attempted to predict whether a user will rate a book above an 8 or below an 8 by using numerical and categorical features. The numerical features we used are: book rate count, book read count, author read count, year of publication, user age, vote score (from the recommender), and matched reader count. The only categorical feature was the book Publishers name.

5.1 Recommendation system

The recommendation algorithm works in two main steps: first narrowing down the massive list of books into a smaller set of candidates, and then predicting which ones should be ranked highest. This candidate generation step relies on what other people with similar reading habits have read. You start by inputting a few seed books, the algorithm then searches the database to find connected users who have also read those exact books, and then it looks at the other titles those connected users have in their libraries. The system assigns a vote score to these new books based on how many seed books the connected user matched with you. The books and vote scores are gathered into a recommendation table and then used to create a subset of the review data only containing those books.

To control how specific these recommendations are, the algorithm uses a statistical rating threshold called k . If we set k to zero (in our analysis, we used $k = 8$), the model just asks what else those people read, resulting in a broad list of thousands of candidate books. However, if we raise the k threshold to eight, the model acts as a stricter filter, only looking at data from users who explicitly gave the seed books a high rating. This reduces the number of matched readers but generates much stronger rating focused recommendations. This model has a few data limitations, for example, splitting reader counts if a single book has multiple ISBN editions.

5.2 Candidate models and pre-processing

Now that we had a way to pick a subset of our dataset and create a derived column, we moved on to the classification task. Our goal was to accurately predict how a user rating of a book

would fall between two distinct classes: rating over 8 (positive class, 1), rating strictly under 8 (negative class, 0). To assist us in this task, we used 5 different classification models, with varying complexities. Initially we used a Logistic Regression (LR) model, but as we tried to improve model performance, we implemented Shallow Tree, Tuned Decision Tree (TDT), Random Forest (RF) and Gradient Boosting (GB). These models try to predict the probability that a row is part of a positive or negative class. Later, we are able to translate the predictions of probability into predictions of 0-1 class labels using a threshold, any probability above the threshold becoming part of the positive class and the opposite for the negative class.

Before we do anything, we must split the data into two sets: testing rows and reporting rows. The testing rows make up 80% of the total data set and are used to train, compare, and score the models, while the other 20%, the model report rows, are used to provide a final analysis of the model performance. In short, testing rows help pick a model, report rows gage the winning models performance in an unbiased way.

Our preprocessing methods involved standardising the numerical features for the logistic model (the other classification models standardise the numerical values on their own), and one-hot encoding of the categorical features for all models.

5.3 Parameter optimization

Because some of the more complex models, namely the TDT, RF and GB, offered a variety of choice in parameters, we decided to perform a grid search in order to find the best ones. This search was conducted using Stratified K-Fold cross validation with the goal of maximising the negative log loss. cross validation was chosen as such because the nature of our research question required us to maintain a proportionally equal amount of the two classes (positive and negative) within each of the cross validation folds. We chose to use negative log loss (maximising negative log loss is the same as minimising log loss; the former term is used because the mathematical algorithms we used in our code always seek to maximise the selected metric) as a performance metric in our grid search because it provides a score based on the probabilities predicted by the model, not the 1-0 labels. Metrics like F1-score first translate the predicted probabilities into positive or negative class labels based on a default threshold of 0.5 (within the sk-learn libraries Grid Search function, otherwise F1 is a metric with a specifiable threshold). This approach would be faulty because model performance depends on the threshold, and the default could be favourable for one set of parameters while unfairly disregarding others.

5.4 Threshold optimization and model scoring

The main scoring method that we used to optimise the threshold and compare the models is F-beta (beta = 0.5). This metric heavily penalises errors such as false positives in a models output in a way that simple F1-scoring was not appropriate for. In the early stages of our modelling, we noticed that trying to maximise F1 would cause our class labels to be almost all positive, with a very small minority being negative. As can be seen in Figure 6, both before and after tuning the threshold, the column corresponding to positive class predictions makes up most of the sample.

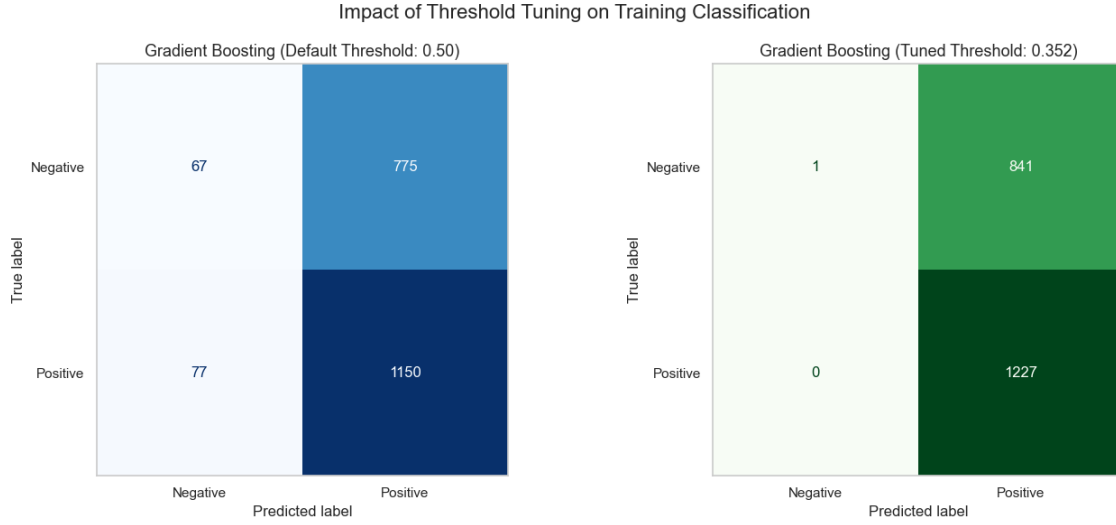


Figure 6: Justification for switching to F-Beta

After establishing optimal parameters where needed and an advantageous scoring metric, we moved on to comparing the models. Using Stratified K-Fold cross validation, we obtained the out-of-fold (OOF) predicted probabilities, assuring that the models get evaluated on unseen data. By varying the threshold from 0% to 100%, classifying the predicted probabilities into 0-1 labels and looking for the highest F-Beta scores, we could then compare all the models, each now equipped with their optimal threshold. As can be seen in Figure 7 and Table 2 all of the models surprisingly seem to score very close to one another. In fact, RF, the highest scoring model with an F-Beta of 0.6379, scores just 0.0053 higher than the lowest scoring one. The optimal thresholds vary between models, but generally remain around the 0.4-0.5 range. RF's threshold is 0.5035, just 0.35% higher than the default threshold.

Model	Optimal Threshold	Peak OOF F05-Score
Random Forest	0.5035	0.6379
Gradient Boosting	0.5199	0.6363
Logistic Regression	0.4750	0.6329
Shallow Tree	0.4128	0.6326
Tuned Decision Tree	0.4128	0.6326

Table 2: Model Performance Metrics

The difference made in optimising the threshold of RF from the default causes a decrease of less than 1% in false positives, as can be seen in the confusion matrices in Figure 8

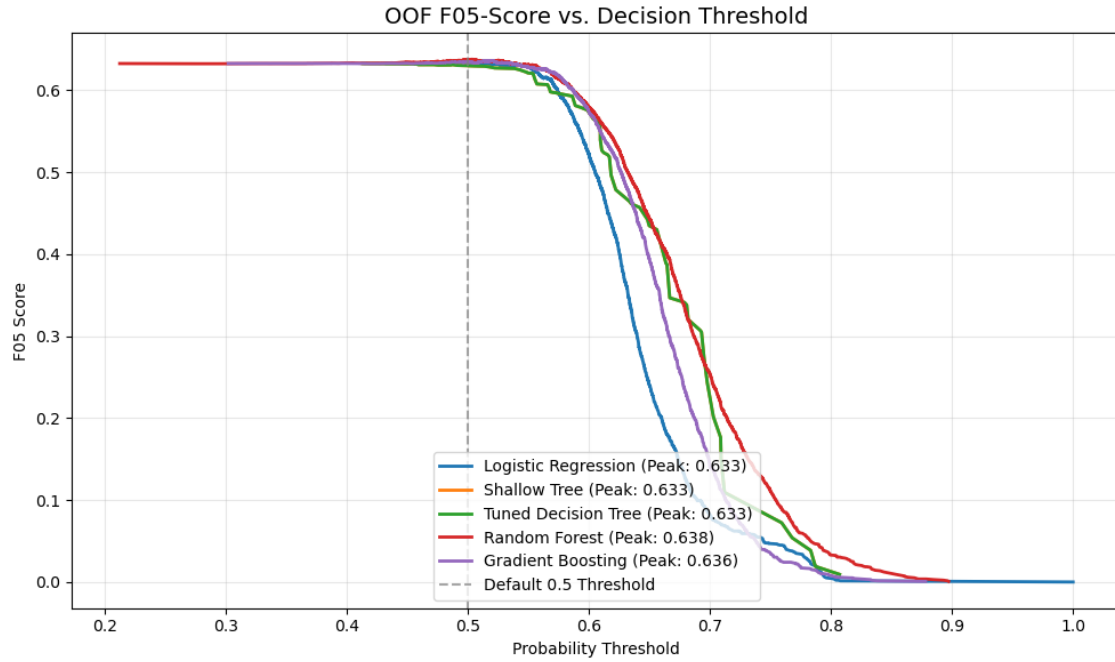


Figure 7: F-Beta scores for different thresholds

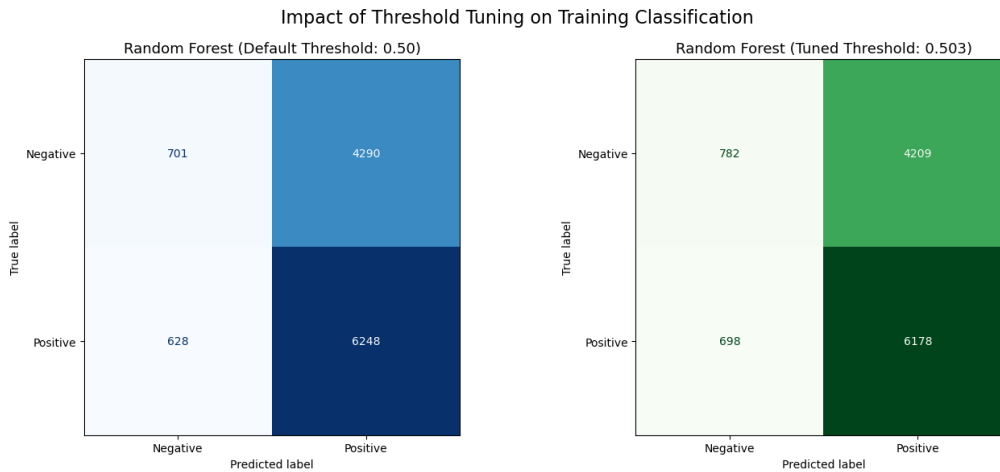


Figure 8: Impact of Threshold Tuning on Training Classification

5.5 Selecting the best model

As further proof of the mathematical conclusion that RF is the best performing model, we can look at the Fold metrics in Figure 9. These two sets of box-plots show the spread of two metrics: F-Beta with the default threshold and negative log loss for the different models. Firstly, note that the Tuned Decision Tree and the Shallow Tree have the exact same metrics because the optimal parameters chosen for the TDT happened to coincide with the shallow tree's parameters. Secondly, we noticed that RF seemed to have a much smaller variance compared to the other models in both metrics, as can be seen by the close grouping of the blue dots corresponding to Random Forrest. However, it is imperative to consider the scale since all points fall within a 0.02 interval for both metrics.

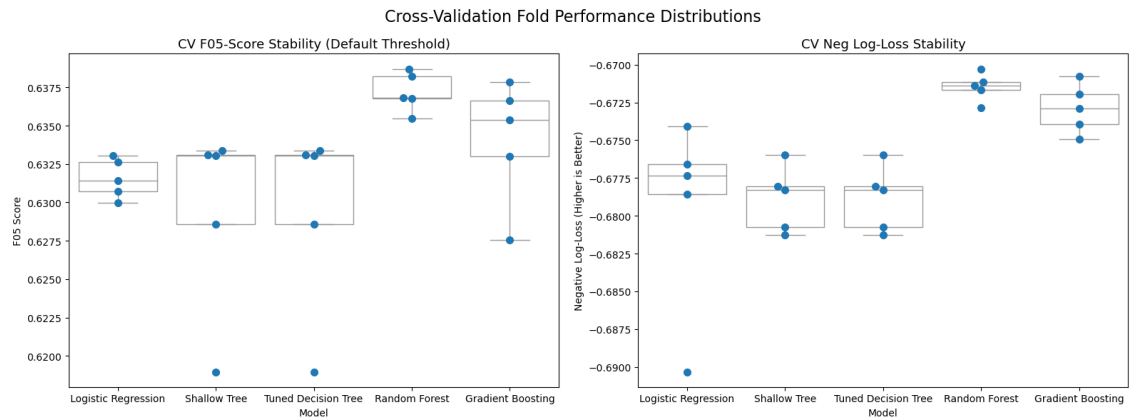


Figure 9: Box-Plots showing fold metrics by model

Figure 10 can help us understand whether a model is over-estimating or underestimating certain probabilities. A model is well calibrated if the corresponding line is very close to the dotted diagonal, while a value much higher than or below the given value exposes some problems. Although most models seem well calibrated on the 0.4 - 0.7 prediction interval, RF seems to be very under-confident for predicted probabilities around 0.25, consistently predicting them lower than they should. On the flip side, TDT and Logistic Regression appear overconfident for probabilities around 0.8 and 0.95 respectively.

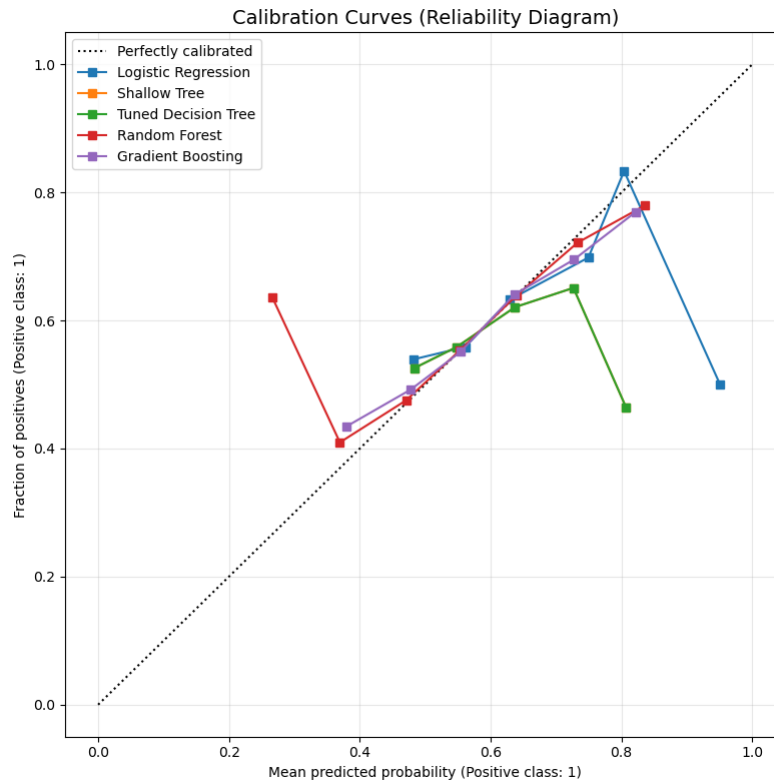


Figure 10: Calibration Curves

5.6 Feature importance

The most influential feature in all models except for LR was by far the age of the user, followed by the year of publication, the author's read count, and the book rate count not necessarily in this order (Figure 11).

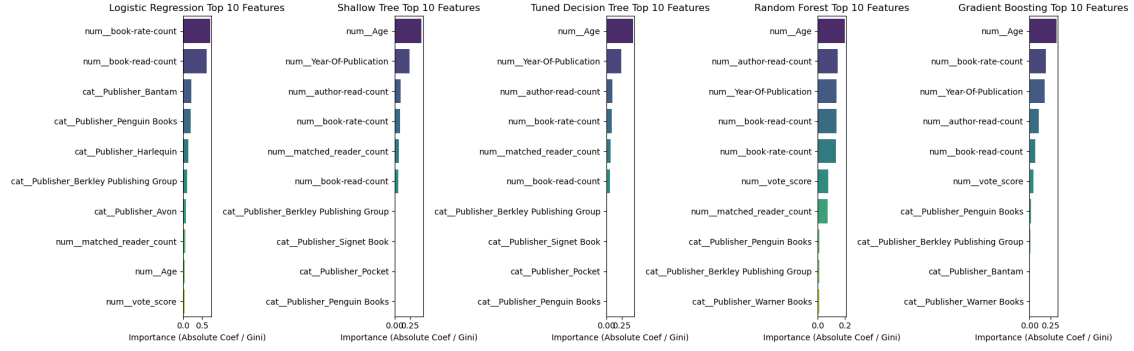


Figure 11: Feature importance Measures

5.7 Validation

Now that we have an optimised model, we can perform some validation on the held out test set. The final accuracy of the model is 59%, which is less than 2% more accurate than if we were to predict that every row was part of the positive class. The precision-recall curve in Figure 12 shows the sacrifice in precision the model makes to predict positive cases well. The recall is 89.8%, which displays the models great ability to detect positive cases, but at a trade off for the precision which sits at a mediocre 59.9%. The confusion matrix in Figure 13 strengthens these findings by showing just how prevalent false positives are, making up 35% of predictions and 40% of positive predictions.

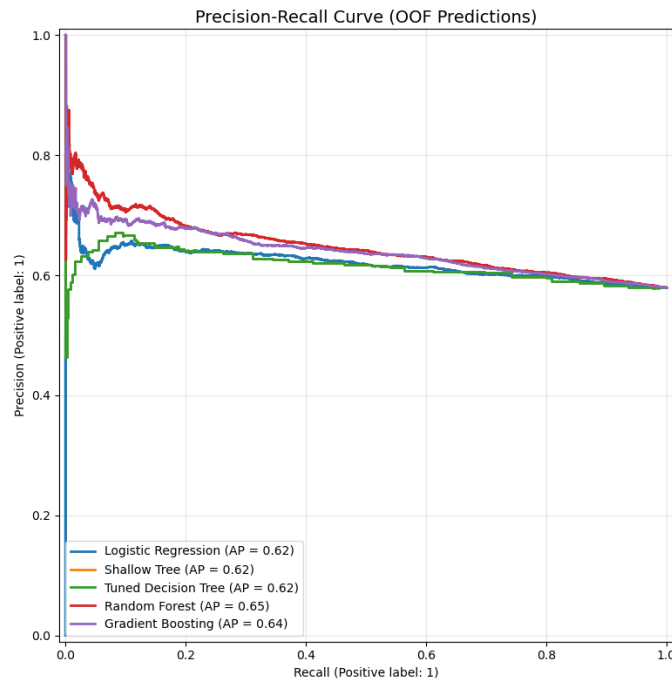


Figure 12: Precision-Recall Curve

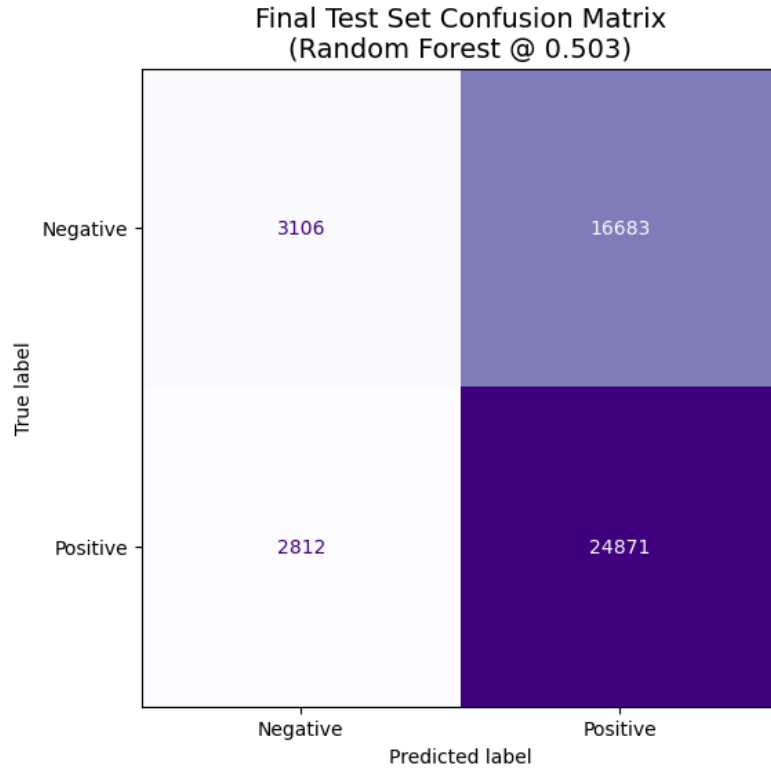


Figure 13: Final Test Confusion Matrix

6 Discussion and Limitations

One limitation is that many ratings in the dataset are recorded as 0. These values do not represent actual rating scores, so they may be better interpreted as implicit interactions. Removing them for prediction reduces the dataset size, while keeping them for recommendation may introduce uncertainty.

Another limitation is missing or unreliable user information. The age variable contains many missing values and the location data required a substantial cleaning. Some users may also provide incorrect ages or incomplete location information, which can affect the analysis of demographic data.

The data set may also be biased because it only includes users and books available in this Kaggle data set. It may not represent all Amazon customers or all books sold on Amazon. Popular books and authors may receive more ratings, while less-known books may be under-represented.

The features of the model are also limited. Important factors such as book genre, review text, user interests, book price, and rating time are not fully included. Because of this, the model may miss important reasons why a user likes or dislikes a book.

7 Conclusion

Overall, this project shows that user reading behavior can be used to create meaningful book recommendations in the Amazon Books dataset. By identifying users who read or highly rated the same seed books, the system can suggest other books with which similar readers also interact. When using a higher rating threshold, such as $k=8$, the recommendations focused more on books that similar users actually liked, rather than just books that were popular or frequently read.

For predicting high ratings, the most important feature is age, other effective features are book-rating-count, publication year, publisher, and author-read-count.

For the prediction task, the results were less successful. The features of book and author popularity, user age, year of publication, and publisher provided some useful signals, but these features were not strong enough to accurately predict whether a user would give a book a high rating. Although Random Forest performed best among the models tested, the difference between the models was small and the final validation performance was only slightly better than a simple baseline. The model achieved high recall, meaning that it could identify many highly rated books, but this came at the cost of low precision and many false positive predictions.

The recommendation system is useful for generating candidate books based on similar reader behavior, but the classification model should not be considered reliable enough to accurately predict high ratings. Future work could improve performance by including stronger features such as book genre, review text, language, rating time, and richer user preference histories. This would help the model capture why users actually like or dislike a book, rather than relying mainly on popularity-based signals.

A Appendix

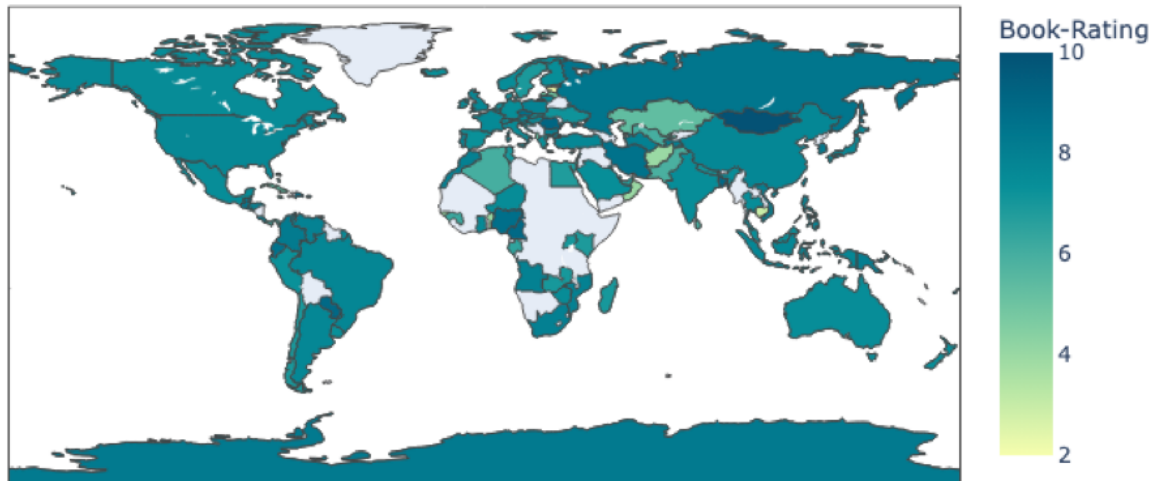


Figure 14: Average Book Rating per Country (all countries included; sample size disregarded; 0-rating (lack of rating) excluded)



Figure 15: Reviews Frequency per Country (Having left a Review)



Figure 16: Reviews Frequency per Country (Not having left a Review)